

Department of Industrial and Systems Engineering and
Engineering Management
The University of Alabama in Huntsville

Description of the Project:

ISE 430/530 Plant Layout

Due Date: Thursday, Nov 20st midnight, Deliverable: One pdf document which incorporates all your figures, tables, etc. into the text. This assignment counts 30-40% of your total grade in the course. You may make use of appendices if it helps with the flow of your document. Your document should be easily readable with few mistakes in grammar and spelling. You should have at least a couple of citations in your document so make sure you do those properly. (See the document that I provided earlier in semester). Make sure you review UAH's policy on plagiarism – I will be asking you to submit your document to Canvas. The minimum penalty for a plagiarism violation on this document is zero on the assignment; the penalty for especially egregious will be an F in the course and filing of academic misconduct charges.

You will need to make and identify many assumptions as part of the design process. You may also need to perform several iterations of the various design process steps. Please document all assumptions and iterations. I am more interested in the process followed than in the final answer obtained. A project that would be considered “very good” will result in a grade in the lower 90%; to achieve a near 100% you must go beyond what is asked and enter into the realm known as “excellent.” I will be looking for innovation and an attention to details. It will be in your best interest to keep your design to yourself (For example if several people have almost the same “good” or “very good” solution I will take this as an average grade and all will receive 70%.- 80%)

I feel like I shouldn't have to say this but you must have all of these items to receive a passing grade. I had a student turn one in without block area design or detailed layout and they argued that that should only count 30 points. YOU MUST do all of these things to get anywhere near a passing grade.

Pink and Green, Inc.

Pink and Green, Incorporated is a well-known contractor to the DOD and is known for their metal fabrication capabilities especially in the area of radar systems. They are expanding their operations and are prepared to open two new facilities to meet expected demand. The two new facilities will be identical. An overall layout is needed in order to begin negotiations with the architectural firm. As industrial engineers for the firm, we are tasked with designing the manufacturing area. Along with these production areas in the facility there are office areas, warehousing, shipping and receiving, and maintenance. We will not be including these in our layout. Management will take our input on these areas and we will be consulted about the relative location of these areas within the overall facility at a later date (after this course!).

Metal Fabrication

There are 10 departments within the metal fabrication area and they are shown in the following table (Table 1). If you have more up to date sizes of machines (from your PL 2 exercise) you may use those instead.

Table 1. Department Information

Department	Abbreviation	Size of Machine in department (machine only – known as footprint)	Number of machines
Milling	M	3'x4'	17
Turning and Boring (lathes)	L	3'x9'	14
Grinding	G	3'x5'	16

Drilling	D	1'x2'	26
Electrical Discharge Machining	EDM	3.5'x6'	14
Electro-Chemical Machining	ECM	3.5'x6.5'	10
CNC Machining Center	MC	8'x9'	16
CNC Turning Center	TC	6'x8'	4
Annealing	A	200 sq.ft. entire area	
Paint	P	250 sq. ft. entire area	

The process plans for hundreds of parts have been analyzed and the sequence of operations can be summarized into the following categories (see Table 2 below). It is believed that these parts are representative of all parts and an estimate of the number of parts per week in each category has been derived based on careful statistical analysis. Note – if the number of loads isn't an integer round up.

Table 2. Part Categories

Product category	Operation sequence	Units/week	Units/load
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1	D-M-ECM-G-A	3000	20
2	TC-L-G-P	2800	20
3	MC-ECM-G-A	2250	30
4	TC-MC-ECM-P	600	10
5	D-M-L-P	3090	55
6	TC-EDM-G-M-A-P	800	80
7	M-EDM-G-A	3625	25
8	TC-L-EDM-P	2200	120
9	TC-L-ECM-G-A	1450	45
10	L-G-ECM-G-P	6500	25
11	MC-M-G-P	1495	65
12	D-G-A-P	2000	30
13	EDM-ECM-D-G-A	3150	25

14	ECM-TC-MC-D	2380	85
15	L-M-D-P	2170	46
16	MC-TC-ECM-A	2100	55

We also know that the ECM should not be located next to the Grinding area.

Provide an analysis for the functional layout of this metal fabrication area. Make any assumptions regarding size and weight of material into area and make sure you state these clearly. Use these assumptions to provide rationale for items such as aisle width, area for WIP, etc. (Failure to state assumptions and the implications of those assumptions will result in a loss of credit for that result. For example, if you use a 10-foot wide aisle you must say why you chose such a width.) All layouts should be done in a drawing package; only initial analysis can be handwritten.

1. Determine the preliminary functional layout of this production area by analyzing material flow between the areas. Provide the appropriate charts used in your analysis. At this stage of the layout the departments should be represented as nodes.
2. Provide detailed workstation layouts for each of the major pieces of production equipment (see Table 1 for size of equipment).
3. Perform some research for aisle widths, etc. Challenge yourself to find different numbers or more detailed information than I gave you in the PowerPoint presentation on SLP.
4. Determine an overall size for each of the departments. In Step 6, you will be asked to provide a detailed layout for one department although you may want to go ahead and do step 6 now so you can make use of that knowledge for the other departments. You do NOT have to provide a detailed layout for all departments, just the one you are assigned in Step 6. In the analysis for this question, you may use a rough idea of the number of machines times the area for each machine and add a percentage for aisle space and other things you'd like in the area. Provide an appropriate block layout that includes the dimensions of all departments including paint and anodize. The desired shape for the total area is rectangular so you may have to increase/decrease computed areas in order to make the final layout fit into this shape. It is permitted to have holes in your design as we will assume those areas will be filled with things like offices, restrooms, cafeteria, etc.
5. Determine the number of forklifts that would be required to service these material handling needs. (Make whatever assumptions are appropriate for this analysis - just make sure you document them.)

6. Develop a detailed layout of the turning/boring department (L) including space for operator, material in and out of machine area, etc.

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Design Project Grade Sheet

I want to emphasize that all of these components are required to receive a passing grade. Recently, a student submitted their work without the block area design or a detailed layout and argued that it should only cost them 30 points. Please understand that you must complete all these elements to achieve anything close to a passing grade.

Your document should be easy to follow and organized, failure to do so may result in an F. I should be able to follow your reasoning and it should be organized according to the topics shown below.

_____ (5) Title Page, Table of Contents, List of Figures

_____ (10) Executive Summary

_____ (15) Metal Fabrication Area Block Design

_____ (15) Research and documentation of aisle widths and area around machines

_____ (15) Detailed Layout of your assigned Department

_____ (20) Analysis of forklift requirements

_____ (10) Referenced Figures, Tables, Appendices

_____ (10) Format and Style (spelling, grammar, neatness)

_____ Out of total 100